

SMART INDIA HACKATHON 2025

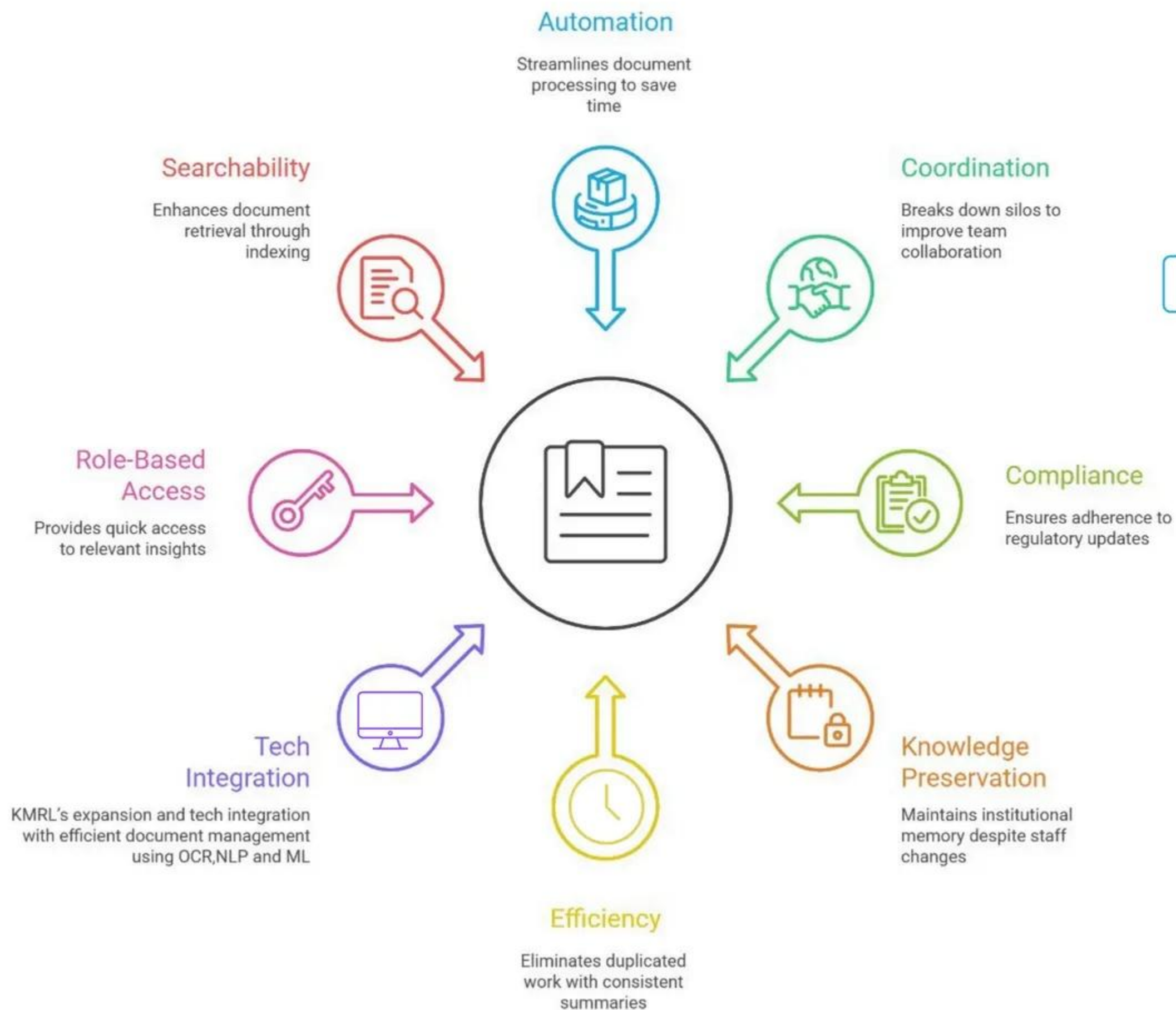
TITLE PAGE

- **Problem Statement ID-** SIH25080
- **Problem Statement Title-** Document Overload at Kochi Metro Rail Limited (KMRL)-An automated solution
- **Theme-** Smart Automation
- **PS Category-** Software
- **Team ID-**
- **Team Name (Registered on portal)-** Framework Fanatics

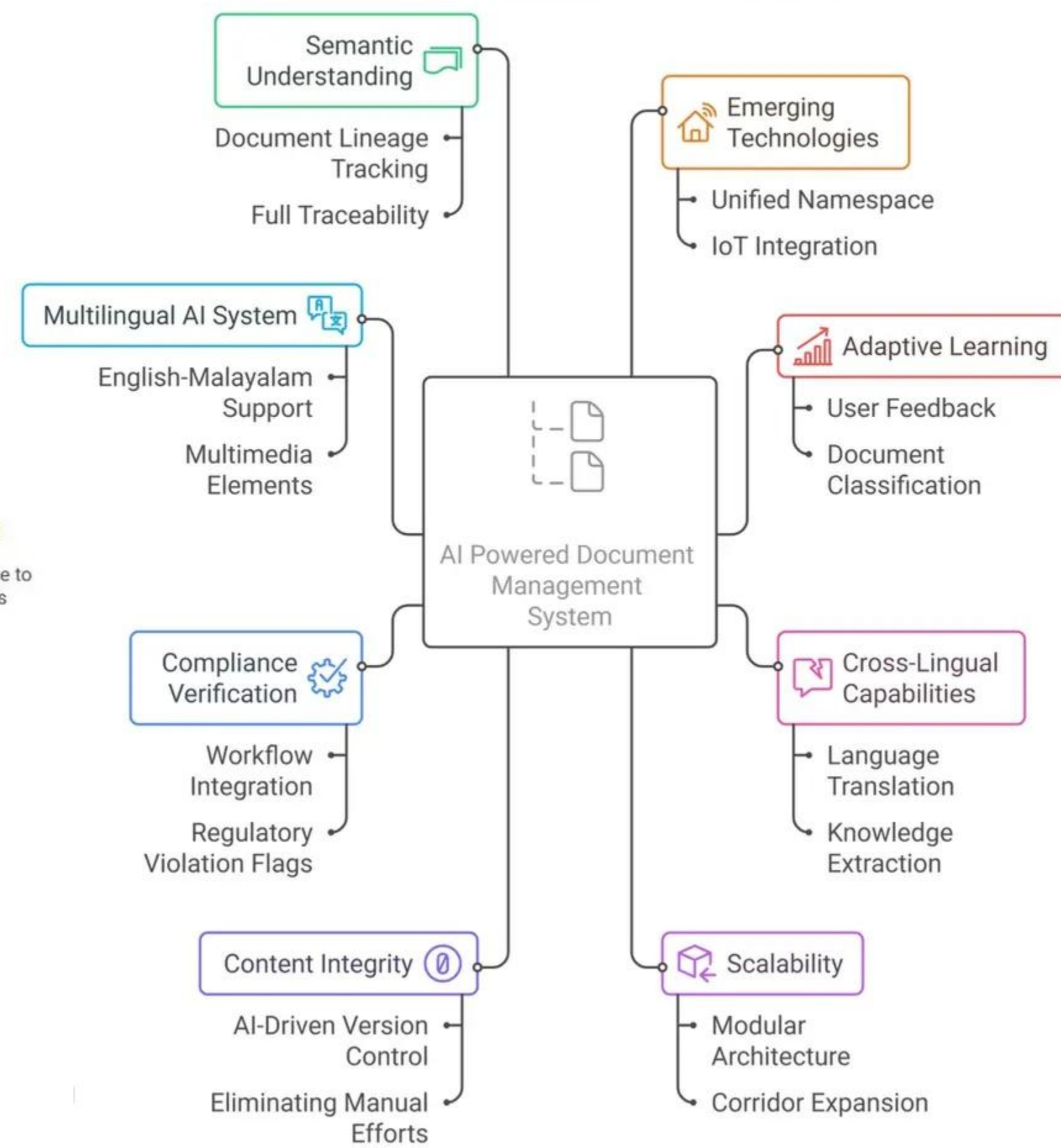


Idea Title

Enhancing KMRL's Document Management



Document Management System: Features and Innovations





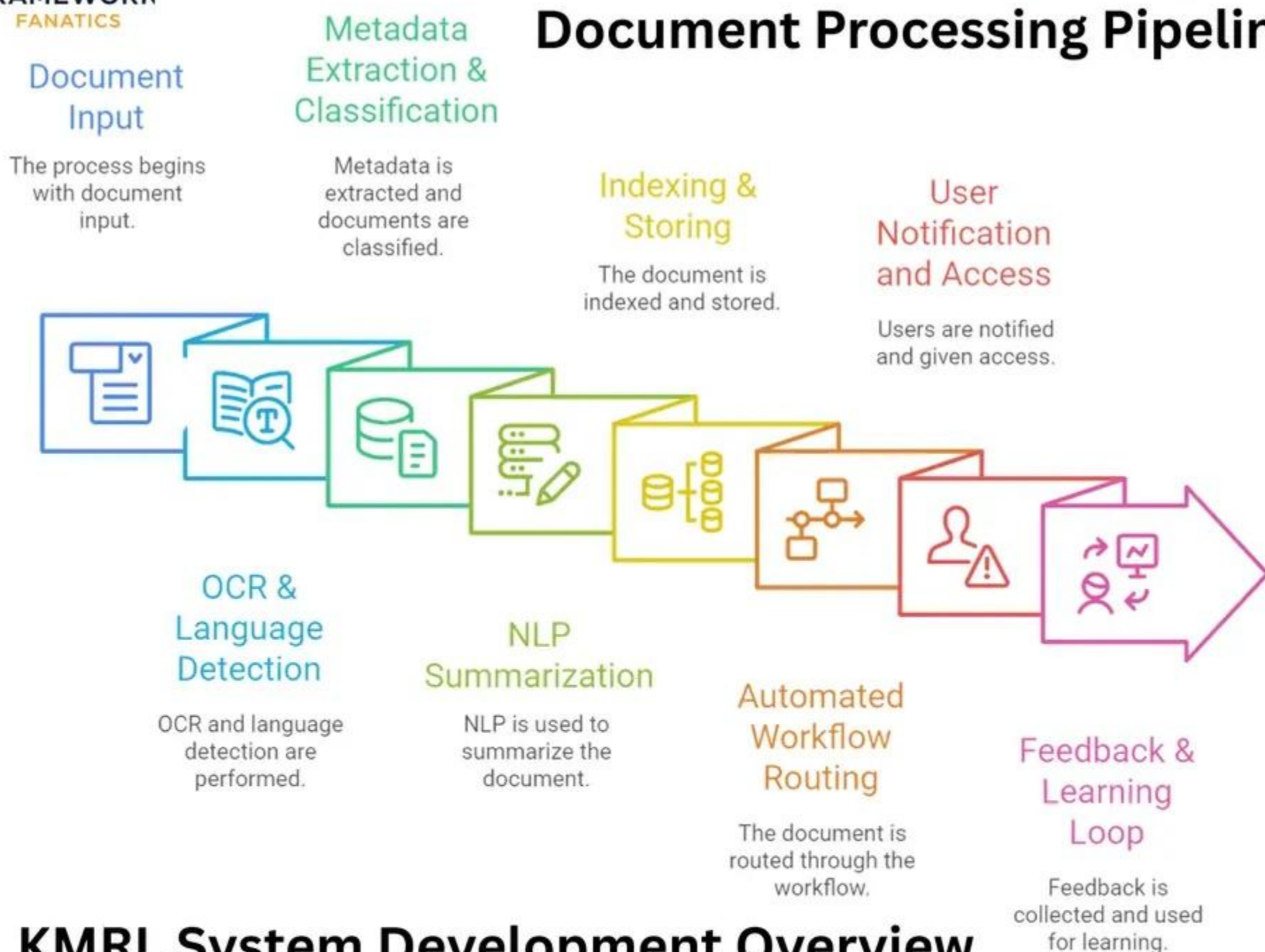
FRAMEWORK
FANATICS

Technical Approach

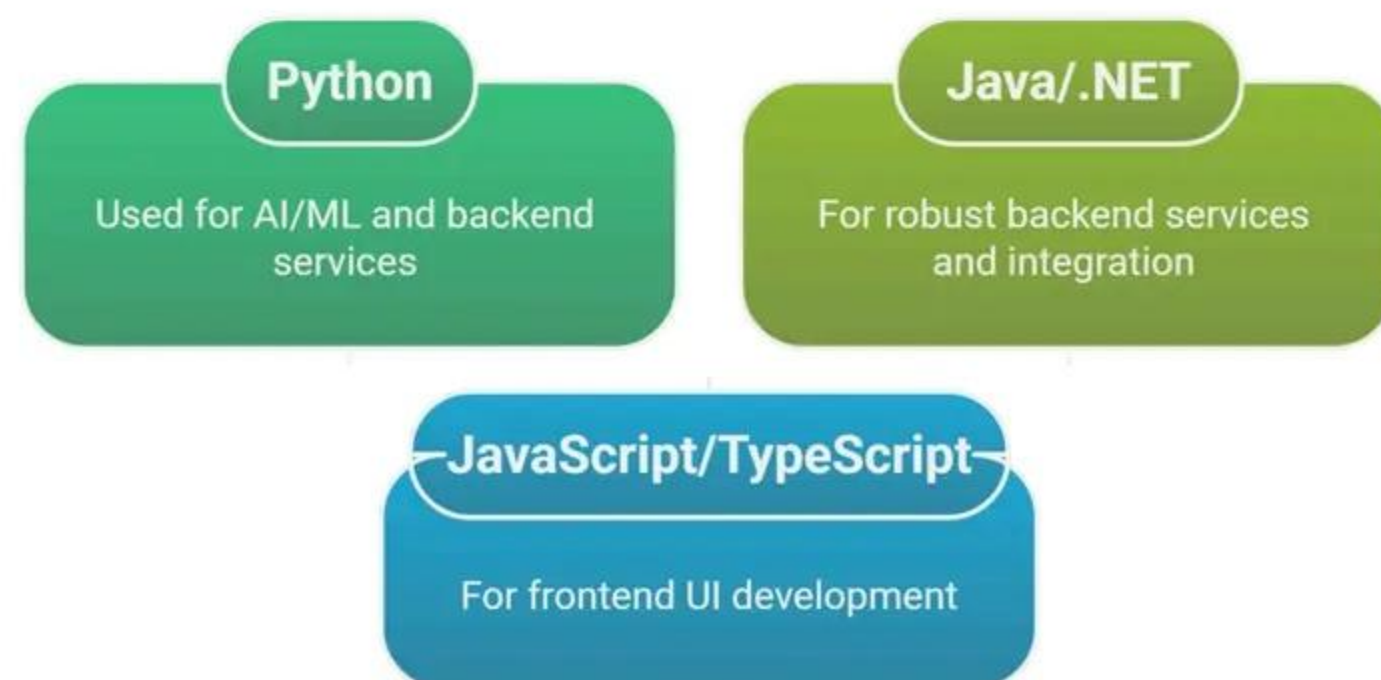


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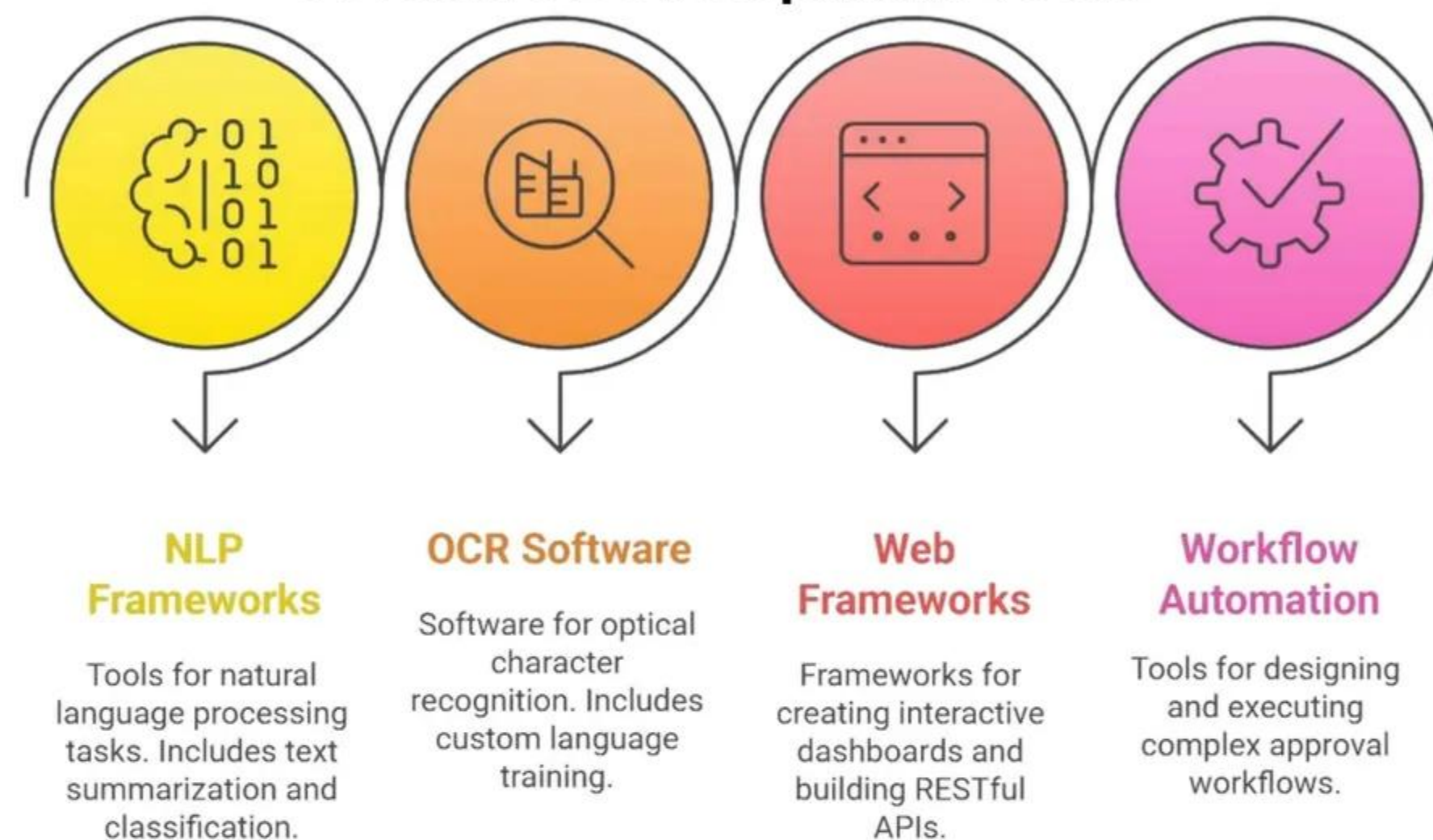
Document Processing Pipelines



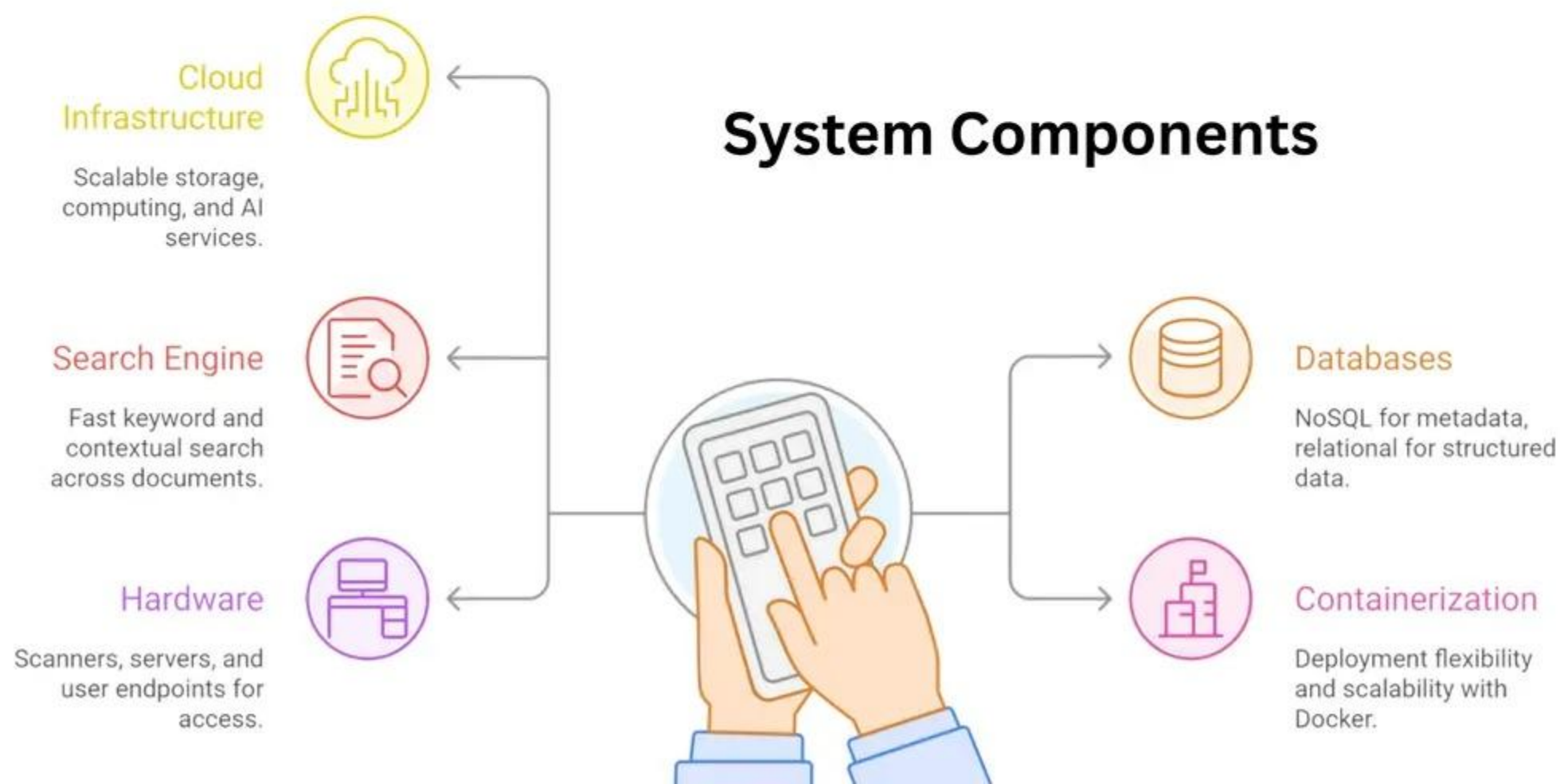
KMRL System Development Overview



Software Development Tools



System Components



Analysis of Feasibility

Technical Feasibility: HIGH

- Proven OCR, NLP, and RPA technologies available
- Existing DDFS infrastructure foundation
- Scalable for diverse document formats and volumes

Financial Feasibility: STRONG ROI

- 70% reduction in document drafting time
- \$35,000 monthly savings (transport authority proven)
- 2,200+ hours saved monthly

Operational Feasibility: EXCELLENT

- Rule-based processes ideal for automation
- Integration with SAP, Maximo, SharePoint systems

Potential Challenges & Risks

Technical Challenges

- Multilingual documents (English/Malayalam)
- Legacy system integration complexity
- Accuracy for handwritten/poor scans

Organizational Risks

- Employee resistance to change
- Training requirements and skill gaps
- Compliance and security concerns

Operational Risks

- System downtime affecting operations
- Data migration challenges

Mitigation Strategies

Technical Solutions

- AI-powered adaptive learning for formats
- Phased rollout (high-impact first)
- Human-in-the-loop validation

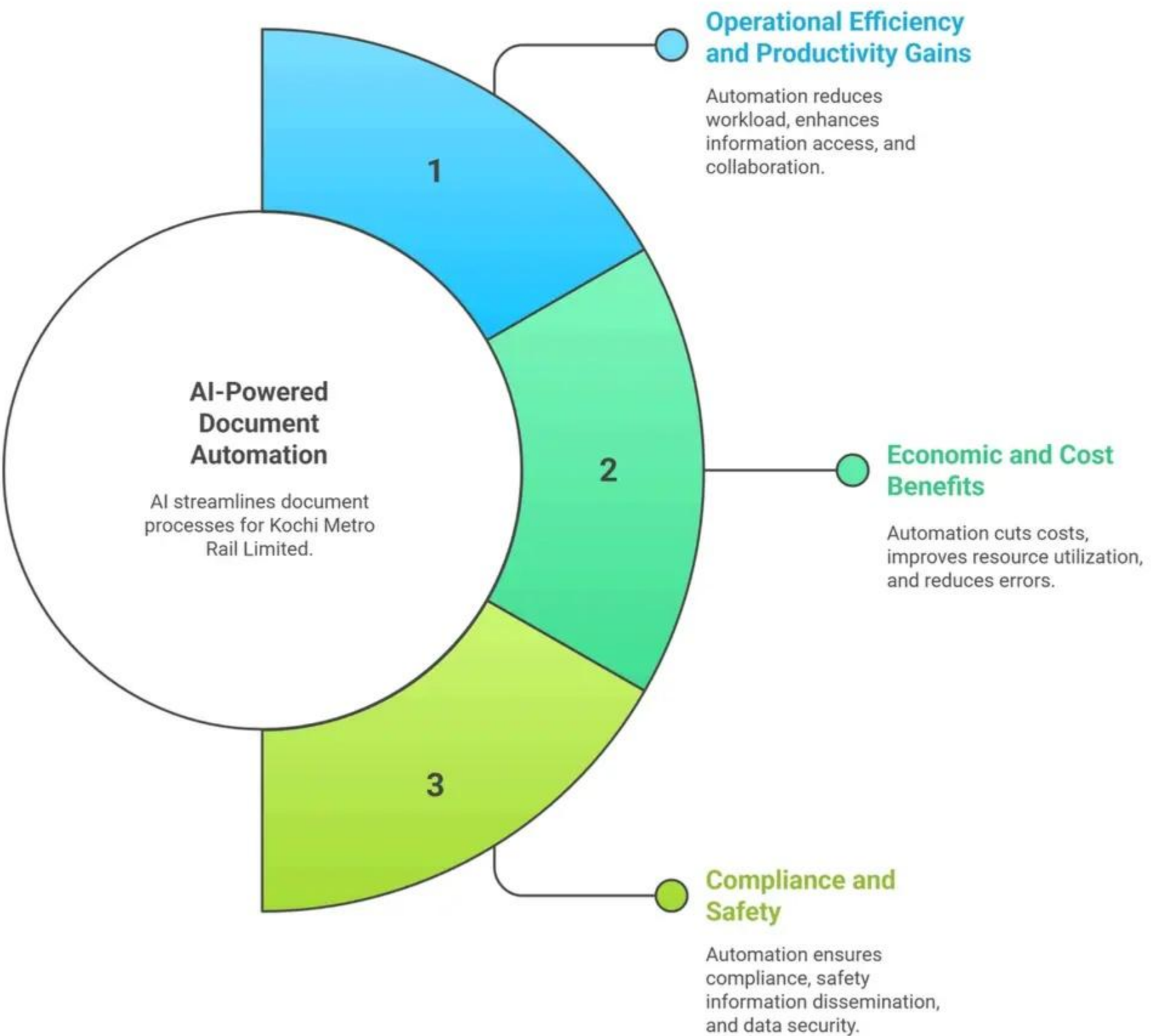
Change Management

- Leadership buy-in & stakeholder engagement
- Comprehensive training programs
- Pilot projects to demonstrate success

Risk Mitigation

- Multi-layered security with audit trails
- Business continuity planning
- Continuous monitoring & optimization

Unveiling The Benefits of AI in Kochi Metro



Key Benefits of Document Automation

Benefit Area	Benefits	Examples
Operational Efficiency	50-70% reduction in manual processing time	Faster train operational decisions
Economic Savings	30-50% reduction in admin labor costs	Lower overhead, redeployed staff
Compliance & Safety	Automated compliance tracking and safety information routing	Reduced audit risks, faster response
Data Security	Role-based access, encrypted sensitive data	Compliance with data protection norms
Social & Passenger Impact	Improved train reliability and service satisfaction	More reliable metro service
Scalability & Future Proofing	Expandable to other networks and integrated with new tech	Support for other metros, IoT

-  [Kochi Metro Rail DPR Phase 1 – Aluva to Petta](#)
-  [Social Due Diligence Report – Kochi Metro Phase 2](#)
-  [Revolutionizing Metro Rail Systems with AI and RDBMS](#)
-  [Document Automation in Mumbai Metro \(eFACiLiTY Case Study\).](#)
-  [Appraisal Guidelines for Metro Rail Project Proposals:](#)
-  [Benchmarking Cost Estimation of Metro Rail Projects](#)
-  [AI applications in transport sector \(Literature Review\).](#)
-  [Metro Rail Asset Performance with IoT, AI, & ML](#)
-  [EIA DUE DILIGENCE REPORT](#)

